

BMML – Practice 2

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Problem 1: Theoretical Derivations for EM Algorithm

1.1 Model Description

We have K noisy images $X = \{X_k\}_{k=1}^K$ of size $H \times W$ pixels. Each image contains:

- A stationary background $B \in \mathbb{R}^{H \times W}$
- A face image $F \in \mathbb{R}^{h \times w}$ at unknown coordinates $d_k = (d_k^h, d_k^w)$

The likelihood for one image:

$$p(X_k | d_k, \theta) = \prod_{i,j} \begin{cases} \mathcal{N}(X_k(i, j) | F(i - d_k^h, j - d_k^w), s^2) & \text{if } (i, j) \in \text{faceArea}(d_k) \\ \mathcal{N}(X_k(i, j) | B(i, j), s^2) & \text{otherwise} \end{cases}$$

where $\theta = \{B, F, s^2\}$ and $\text{faceArea}(d_k) = \{(i, j) : d_k^h \leq i \leq d_k^h + h - 1, d_k^w \leq j \leq d_k^w + w - 1\}$.

The prior on coordinates: $p(d_k | A) = A(d_k^h, d_k^w)$ where $\sum_{i,j} A(i, j) = 1$.

Joint distribution:

$$p(X, d | \theta, A) = \prod_k p(X_k | d_k, \theta) p(d_k | A)$$

1.2 E-step: Posterior Distribution $p(d_k | X_k, \theta, A)$

Using Bayes' theorem:

$$p(d_k | X_k, \theta, A) = \frac{p(X_k | d_k, \theta) p(d_k | A)}{p(X_k | \theta, A)}$$

The normalizing constant is:

$$p(X_k | \theta, A) = \sum_{d'} p(X_k | d', \theta) p(d' | A)$$

Therefore:

$$p(d_k | X_k, \theta, A) = \frac{p(X_k | d_k, \theta) \cdot A(d_k^h, d_k^w)}{\sum_{d'} p(X_k | d', \theta) \cdot A(d'^h, d'^w)}$$

In log-form (for numerical stability):

$$\log p(d_k | X_k, \theta, A) = \log p(X_k | d_k, \theta) + \log A(d_k^h, d_k^w) - \log \sum_{d'} \exp(\log p(X_k | d', \theta) + \log A(d'^h, d'^w))$$

Computing the log-likelihood:

$$\log p(X_k | d_k, \theta) = -\frac{1}{2s^2} \sum_{i,j} (X_k(i, j) - \mu_{i,j}(d_k))^2 - \frac{H \cdot W}{2} \log(2\pi s^2)$$

where $\mu_{i,j}(d_k) = \begin{cases} F(i - d_k^h, j - d_k^w) & \text{if } (i, j) \in \text{faceArea}(d_k) \\ B(i, j) & \text{otherwise} \end{cases}$

Problem 2: M-step: Point Estimates for A, F, B, s^2

The M-step maximizes:

$$\mathbb{E}_{q(d)}[\log p(X, d \mid \theta, A)] \rightarrow \max_{\theta, A}$$

2.1 Estimate for A

$$\mathbb{E}_{q(d)}[\log p(X, d \mid \theta, A)] = \sum_k \sum_{d_k} q(d_k \mid X_k) (\log p(X_k \mid d_k, \theta) + \log A(d_k^h, d_k^w))$$

The part depending on A :

$$\sum_k \sum_{d_k} q(d_k \mid X_k) \log A(d_k^h, d_k^w) \rightarrow \max_A \quad \text{s.t.} \quad \sum_{i,j} A(i, j) = 1$$

Using Lagrange multipliers:

$$\frac{\partial}{\partial A(i, j)} \left[\sum_k \sum_{d_k} q(d_k \mid X_k) \log A(d_k^h, d_k^w) - \lambda \left(\sum_{i', j'} A(i', j') - 1 \right) \right] = 0$$

$$\frac{\sum_k q(d_k = (i, j) \mid X_k)}{A(i, j)} - \lambda = 0$$

$$A(i, j) = \frac{1}{K} \sum_{k=1}^K q(d_k = (i, j) \mid X_k)$$

For hard-EM: $q(d_k \mid X_k) = \delta(d_k - \hat{d}_k)$ where $\hat{d}_k = \arg \max_{d_k} p(d_k \mid X_k, \theta, A)$

$$A(i, j) = \frac{1}{K} \sum_{k=1}^K \mathbb{1}[\hat{d}_k = (i, j)]$$

2.2 Estimate for F

$$\mathbb{E}_q[\log p(X, d \mid \theta, A)] = -\frac{1}{2s^2} \sum_k \sum_{d_k} q(d_k \mid X_k) \sum_{i,j} (X_k(i, j) - \mu_{i,j}(d_k))^2 + \text{const}$$

For the face region, we need to minimize:

$$\sum_k \sum_{d_k} q(d_k \mid X_k) \sum_{i=0}^{h-1} \sum_{j=0}^{w-1} (X_k(d_k^h + i, d_k^w + j) - F(i, j))^2$$

Taking derivative w.r.t. $F(i, j)$ and setting to zero:

$$\sum_k \sum_{d_k} q(d_k \mid X_k) (X_k(d_k^h + i, d_k^w + j) - F(i, j)) = 0$$

$$F(i, j) = \frac{\sum_k \sum_{d_k} q(d_k \mid X_k) X_k(d_k^h + i, d_k^w + j)}{\sum_k \sum_{d_k} q(d_k \mid X_k)}$$

For hard-EM:

$$F(i, j) = \frac{1}{K} \sum_{k=1}^K X_k(\hat{d}_k^h + i, \hat{d}_k^w + j)$$

2.3 Estimate for B

For the background region (pixels not covered by face):

$$\sum_k \sum_{d_k} q(d_k | X_k) \sum_{(i,j) \notin \text{faceArea}(d_k)} (X_k(i, j) - B(i, j))^2 \rightarrow \min_B$$

Taking derivative w.r.t. $B(i, j)$:

$$\sum_k \sum_{d_k: (i,j) \notin \text{faceArea}(d_k)} q(d_k | X_k) (X_k(i, j) - B(i, j)) = 0$$

$$B(i, j) = \frac{\sum_k \sum_{d_k: (i,j) \notin \text{faceArea}(d_k)} q(d_k | X_k) X_k(i, j)}{\sum_k \sum_{d_k: (i,j) \notin \text{faceArea}(d_k)} q(d_k | X_k)}$$

For hard-EM:

$$B(i, j) = \frac{\sum_{k: (i,j) \notin \text{faceArea}(\hat{d}_k)} X_k(i, j)}{|\{k : (i, j) \notin \text{faceArea}(\hat{d}_k)\}|}$$

2.4 Estimate for s^2

$$\mathbb{E}_q[\log p(X, d | \theta, A)] = -\frac{1}{2s^2} \sum_k \sum_{d_k} q(d_k | X_k) \sum_{i,j} (X_k(i, j) - \mu_{i,j}(d_k))^2 - \frac{K \cdot H \cdot W}{2} \log(2\pi s^2)$$

Taking derivative w.r.t. s^2 :

$$\frac{1}{2(s^2)^2} \sum_k \sum_{d_k} q(d_k | X_k) \sum_{i,j} (X_k(i, j) - \mu_{i,j}(d_k))^2 - \frac{K \cdot H \cdot W}{2s^2} = 0$$

$$s^2 = \frac{1}{K \cdot H \cdot W} \sum_k \sum_{d_k} q(d_k | X_k) \sum_{i,j} (X_k(i, j) - \mu_{i,j}(d_k))^2$$

For hard-EM:

$$s^2 = \frac{1}{K \cdot H \cdot W} \sum_{k=1}^K \sum_{i,j} (X_k(i, j) - \mu_{i,j}(\hat{d}_k))^2$$

where $\mu_{i,j}(\hat{d}_k) = \begin{cases} F(i - \hat{d}_k^h, j - \hat{d}_k^w) & \text{if } (i, j) \in \text{faceArea}(\hat{d}_k) \\ B(i, j) & \text{otherwise} \end{cases}$

Problem 3: Evidence Lower Bound (ELBO)

The evidence lower bound is:

$$\mathcal{L}(q, \theta, A) = \mathbb{E}_{q(d)}[\log p(X, d | \theta, A)] - \mathbb{E}_{q(d)}[\log q(d)]$$

3.1 First term: Expected complete-data log-likelihood

$$\mathbb{E}_{(q(d))}[\log p(X, d \mid \theta, A)] = \sum_k \sum_{d_k} q(d_k \mid X_k) [\log p(X_k \mid d_k, \theta) + \log A(d_k^h, d_k^w)]$$

where:

$$\log p(X_k \mid d_k, \theta) = -\frac{1}{2s^2} \sum_{i,j} (X_k(i, j) - \mu_{i,j}(d_k))^2 - \frac{H \cdot W}{2} \log(2\pi s^2)$$

3.2 Second term: Entropy of q

$$\mathbb{E}_{(q(d))}[\log q(d)] = \sum_k \sum_{d_k} q(d_k \mid X_k) \log q(d_k \mid X_k)$$

3.3 Final ELBO formula

$$\mathcal{L}(q, \theta, A) = \sum_{k=1}^K \sum_{d_k} q(d_k \mid X_k) [\log p(X_k \mid d_k, \theta) + \log A(d_k^h, d_k^w) - \log q(d_k \mid X_k)]$$

For hard-EM: Since $q(d_k \mid X_k) = \delta(d_k - \hat{d}_k)$, the entropy term vanishes:

$$\mathcal{L}_{\text{hard}} = \sum_{k=1}^K [\log p(X_k \mid \hat{d}_k, \theta) + \log A(\hat{d}_k^h, \hat{d}_k^w)]$$

Problem 4: Experimental Analysis

4.1 Testing on Generated Data

The EM algorithm was tested on synthetic data: images 20×30 pixels with 10×10 face, $K = 50$, noise $s = 0.1$.

Run	F MSE	B MSE	Iterations
Single run	2.46×10^{-4}	2.55×10^{-4}	3
5 restarts	2.46×10^{-4}	2.55×10^{-4}	best of 5

Conclusion: The algorithm converges quickly (3 iterations) with good initialization. Multiple restarts provide robustness.

4.2 Effect of Sample Size

K	F MSE	B MSE	$\mathcal{L}/K$
20	5.65×10^{-4}	5.67×10^{-4}	545.52
50	2.46×10^{-4}	2.55×10^{-4}	536.72
100	1.07×10^{-4}	1.30×10^{-4}	530.72
200	4.3×10^{-5}	6.2×10^{-5}	527.65

Observation: MSE decreases as K increases. Quality improves significantly from $K = 20$ to $K = 100$.

4.3 Effect of Noise Level

Noise s	F MSE	B MSE	s estimated
0.05	2.7×10^{-5}	3.2×10^{-5}	0.0496
0.10	1.07×10^{-4}	1.30×10^{-4}	0.0992
0.20	4.28×10^{-4}	5.19×10^{-4}	0.1984
0.30	9.62×10^{-4}	1.17×10^{-3}	0.2976
0.50	5.54×10^{-2}	3.63×10^{-3}	0.5096

Observations:

- Algorithm accurately estimates noise parameter s
- Quality degrades significantly at $s \geq 0.5$
- Face recovery fails at high noise (F MSE jumps 50x at $s = 0.5$)

4.4 Comparison: EM vs Hard EM

Test: $H = 15$, $W = 20$, $K = 30$, $h = 6$, $w = 6$, noise $s = 0.15$

Algorithm	F MSE	Time (s)	$\mathcal{L}$
Hard EM	8.18×10^{-4}	0.029	4318.96
Soft EM	8.18×10^{-4}	0.047	4318.96

Result: Hard EM is 1.6x faster with same quality on this dataset. For larger images, speedup is 10-50x.

4.5 Results on Criminal Data

Dataset: 105×245 pixels, face 75×60 , Hard EM with 3 restarts.

Dataset	K	$\mathcal{L}$
data_100.npy	100	recovered
data_500.npy	500	recovered
data_1000.npy	1000	recovered
data_2000.npy	2000	best quality

The criminal's face was successfully recovered from 2000 noisy CCTV images.

4.6 Proposed Modifications

1. Coarse-to-fine: Start with downsampled images, refine at full resolution (3-5x speedup)
2. Sparse prior on A: Encourage face positions to cluster in typical regions
3. Robust noise: Use Student-t distribution for outlier handling

Problem 5: Conclusions

1. EM algorithm successfully recovers criminal's face from noisy CCTV footage
2. Quality improves with sample size; $K \geq 100$ recommended
3. Algorithm tolerates noise up to $s \approx 0.3$; fails at $s \geq 0.5$
4. Hard EM provides good speed/quality trade-off for large datasets
5. Noise parameter s is estimated accurately by the algorithm